

# TRAINED AGENTIC CONTEXT MANAGEMENT

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## ABSTRACT

We study long context language models. Instead of training long context natively, or designing a long context harness, we train a model over the simplest possible harness: a tool to call itself with any specified prompt and a tool to read tokens in a range from the input context. We finetune Qwen3.6-35B-A3B on a diverse synthetic dataset using this harness. With only 8,000 tokens of context, our small model is as strong as GPT-5.4 with 1M tokens of context on the OOLONG-synth benchmark when document length exceeds 40K tokens<sup>1</sup>.

## 1 INTRODUCTION

As language models boast impressive breakthroughs (Luong & Lockhart, 2025; Wei, 2025; OpenAI, 2025b; Lin & Cheng, 2025; OpenAI, 2026g,f,e), increasing focus is placed on their limitations. The most salient resource constraint of a language model is its context length. Models can process and produce tokens within this limit, but the limit is hard. Model producers have identified this constraint and placed great emphasis on extending it as large as possible, with some model builders touting 10M (Dangel, 2026; Meta AI, 2025; Zhu et al., 2026) and even 100M (Magic, 2024) token context lengths (for reference, the length of the first Harry Potter book is about 100,000 tokens, the full U.S. tax code roughly 3M, and the Linux kernel source code about 300M).

While native context scaling is appealing as a concrete engineering problem, it is not the only approach to producing a model capable of processing long documents. In 2021, OpenAI fine-tuned GPT-3 to summarize sections of books, and then summarize those summaries, achieving competitive performance with human book summaries on a model with a context length of just 2,048 tokens (Wu et al., 2021). As the field has matured, agentic harnesses have become viable. MemGPT treats context similarly to an operating system, relying on LLM-invoked function calls to push relevant information in and out of the context window to external memory, effectively achieving expanded virtual context (Packer et al., 2023). More recently, Recursive Language Models introduced a harness that utilizes LLMs’ coding prior to call itself in a REPL environment, achieving stronger performance than base models on retrieval and aggregation tasks both within context limits, but also beyond it, expanding effective context by  $\sim 4\times$  (Zhang et al., 2025).

While handcrafted prompts and programmatic environments can be effective in solving long context problems without native long context training, it begs an obvious question: if models can, for example, solve all the problems at the International Collegiate Programming Contest<sup>2</sup> World Finals (OpenAI, 2025b), or more generally, implement requested features in codebases far larger than context limits (Marten et al., 2026; OpenAI, 2026d; Anthropic, 2026b), can they not be *trained* to manage context effectively, *without human-authored harness crutches*? Such model training internalizes decisions external harnesses hardcode, offering the potential to scale much more generally.

We answer this question in this paper, affirmatively. We consider the simplest possible agentic context harness, defined below. The purpose of the harness is to *enable capability*, without *fixing choices* or *instilling a prior*.

<sup>1</sup>Our code is available at: <https://github.com/brycesandlund/infinite-context>.

<sup>2</sup><https://icpc.global/>

## 1.1 HARNESS

Our harness has two tools:

1. `read_chunk(start, end)`: Read a slice of the document and return the decoded text of tokens `[start, end)`.
2. `spawn_subagent(subtask)`: Spawn a fresh-context copy of yourself to solve `subtask`; returns the child's `\boxed{}` answer.

Unmodified long-context prompts are provided to the model over a document accessed via the above harness. The system prompt is propagated to child agents and includes the above harness definition, a generic instruction, and potentially task-specific background, in accordance with the respective eval<sup>3</sup>. Subagents' user prompt is specified by the calling model.

## 1.2 RESULTS

We finetune Qwen3.6-35B-A3B on a diverse synthetic dataset using this harness and achieve the following results:

1. On the OOLONG-synth benchmark, with a context limit of just 8K tokens per agent, our small model performs **as well** as GPT-5.4 with 1M tokens of context when document length exceeds 40K tokens.
2. On the RULER benchmark, with a context limit of just 8K tokens per agent, our model scores  $\geq 85\%$  on document size from 10K to 320K tokens.

## 1.3 PAPER ORGANIZATION

In the next section, we survey related work to frame the contribution of the following sections. In Section 3, we describe our training methodology. In Section 4, we discuss our experiments, and we conclude in Section 5.

# 2 RELATED WORK

Long-context LLM work can be broken into a few distinct categories: *native long context models*, *context compaction*, and *long context harnesses*, both trained and untrained.

## 2.1 NATIVE LONG CONTEXT MODELS

GPT-2 shipped with a 1,024 token context limit (Radford et al., 2019). This limit doubled in every generation through GPT-4 (Brown et al., 2020; OpenAI, 2022; 2023b), until jumping to the 100K range on GPT-4 Turbo (OpenAI, 2023a), with contemporaneous models Claude 2 (Anthropic, 2023) and later Llama 3.1 (Grattafiori et al., 2024) at similar limits. The core difficulty in scaling context length  $n$  in the classic Transformer architecture (Vaswani et al., 2017) is  $\Omega(n^2)$  attention complexity, which becomes prohibitively expensive as  $n \gg 100K$ . State-space architectures like the mamba lineage (Gu & Dao, 2024; Dao & Gu, 2024; Lahoti et al., 2026), neural methods like Titans (Behrouz et al., 2024), and Linear Transformer (Katharopoulos et al., 2020) derivatives DeltaNet (Schlag et al., 2021) and Gated DeltaNet (Yang et al., 2025) abandon quadratic attention for recurrence. Modern architectures are hybrid, combining linear or sparse attention layers with dense attention, allowing context lengths to scale to 1M+ tokens while retaining long-context performance (Qwen Team, 2026; Kimi Team, 2026b; DeepSeek-AI, 2025; GLM-5 Team, 2026; NVIDIA, 2025; Gemma Team, 2024; Mistral AI, 2024).

## 2.2 CONTEXT COMPACTION

An alternative to native long context, and an inevitability when such native context fills, is a function  $f$  mapping current context  $x$  to a reduced context  $y$  where  $|y| \ll |x|$ . Three categories of

<sup>3</sup>Precise harness environment is given in Appendix A.

such lossy  $f$  exist: *masking*, *summarization*, and *learned compaction*. The simplest compaction technique is masking: dropping tokens based on a predetermined rule. Clearing tool call results is a popular recommended choice (Anthropic, 2025a), achieving results competitive with summarization (DeepSeek-AI, 2025; Lindenbauer et al., 2025). Alternative approaches evict tokens based on attention statistics (Xiao et al., 2024; Zhang et al., 2023; Li et al., 2024). Summarization is the current production harness default, implemented in Claude Code, Codex, and Gemini CLI (Anthropic, 2025a; OpenAI, 2026a; Google, 2026; Barbaste et al., 2026). Trained compaction is a new frontier, described as part of GPT-5.1-Codex-Max (OpenAI, 2025a) and Cursor’s composer (Cassano & Rush, 2026) training. “Still” trains a model to compress KV-Cache directly (O’Neill et al., 2026).

### 2.3 LONG CONTEXT HARNESSES

As long as LLMs have proven useful, an evolving line of work has focused on how to expand their capabilities without touching model internals. The first efforts were known as prompt engineering (Phoenix & Taylor, 2024; Berryman & Ziegler, 2024), which, though necessary on pre-trained base models (Brown et al., 2020), became internalized through instruction-tuning and reasoning RL (Ouyang et al., 2022; Guo et al., 2025). Function-calling has led to a new wave of such efforts, where domain-specific knowledge, tools, and multi-agent systems live on a scaffold that interacts with powerful LLMs known as a “harness” (Lopopolo, 2026; Barbaste et al., 2026).

Long-context harnesses originate with systems like MemGPT (Packer et al., 2023), RAPTOR (Sarathi et al., 2024), and Chain of Agents (Zhang et al., 2024), where the overall goal is to engineer workflows that decompose documents into contexts independent LLM calls can read and aggregate. Recursive Language Models is a popular recent example that allows for single-level recursive LLM calls via a programmatic environment (Zhang et al., 2025). Follow-up work expands subagent capabilities (Lumer et al., 2026), trains shared parent/child policies (Kim & Ahmad, 2026), and incorporates self-reflection (Alizadeh et al., 2026).

Memagent (Yu et al., 2026) trains a harness that walks linearly through context, using RL and a harness prompt to determine what to carry from node-to-node. Similarly to our work, they also use a small (8K token) context while processing long (as much as 3.5M token) documents. However, they test primarily on the distribution they train, compare only against other small models, and hardcode execution path and prompts in the harness.

Frontier models now utilize multi-agent systems. Kimi K2.5 trains “agent swarm” (Kimi Team, 2026a): the model is trained to call subagents via reinforcement learning. Subagent trajectories are not trained. GPT-5.6 has also trained multi-agent behavior (OpenAI, 2026c), notably utilizing 10,000 agents to solve Navier-Stokes (OpenAI, 2026e). Anthropic, DeepMind, Meta, and xAI likely do similarly, with multi-agent systems touted in respective model and product releases (Anthropic, 2025b; 2026a; OpenAI, 2026b; xAI, 2026; 2025; Meta Superintelligence Labs, 2026; Google, 2025; Gottweis et al., 2026).

## 3 METHODS

We deliberately use the minimal tool definitions necessary to train agentic context management. However, our harness is sufficient to allow arbitrarily complex multi-agent decompositions. Decompositions in the literature fall into roughly three categories: *tree reduce* (Wu et al., 2021; Sarathi et al., 2024; Zhang et al., 2022), *sequential folds* (Yu et al., 2026; Zhang et al., 2024), and *agentic delegation* (Zhang et al., 2025; Kimi Team, 2026a; Anthropic, 2025b). We utilize all three strategies.

Without any training, models generally make poor use of our harness. Base Qwen3.6-35B-A3B, GPT-5.4, and Kimi K2.6 use subagent delegation sporadically, sometimes choosing to read the entire document in one context, even if context limit is specified to be less than document length, and other times delegating the exact original question to subagents. Opus 4.6 does slightly better, showing real context pressure awareness and document range subagent delegation. Unfortunately it delegates *too* much, blowing through \$300 of API credits building massive subagent trees on a 100-question eval,

only to overcount<sup>4</sup>. Instead, we compare to base models supplied the entire document in context with native context limits, where performance is stronger.

In general, models have a *weak prior* for context decomposition. Pre-training is pure language continuation, and agentic post-training does not teach decomposition strategies. Where Recursive Language Models (Zhang et al., 2025) can rely on strong coding priors instilled in both pre- and post-training, decomposition on our harness needs to be taught.

One idea might be to RL an open weight model on our harness. Unfortunately, exploration in LLM RL is far too weak to escape the weak decomposition prior at modest compute budgets. It is also unclear what reward to assign subagents, and if GRPO (Shao et al., 2024) or its derivatives (Yu et al., 2025; Liu et al., 2025; Zheng et al., 2025) are used, what group to compare subagent trajectories in.

We instead SFT a decomposition strategy. We train *tree reduce* for order-independent tasks (see Figure 1), *sequential fold* for order-dependent tasks (see Figure 2), and *agentic delegation* to handle boundary conditions whenever we read the document (see Figure 3). By not hardcoding a decomposition structure in our harness, our model internalizes and composes delegation strategies according to the task and current problem-solving state.

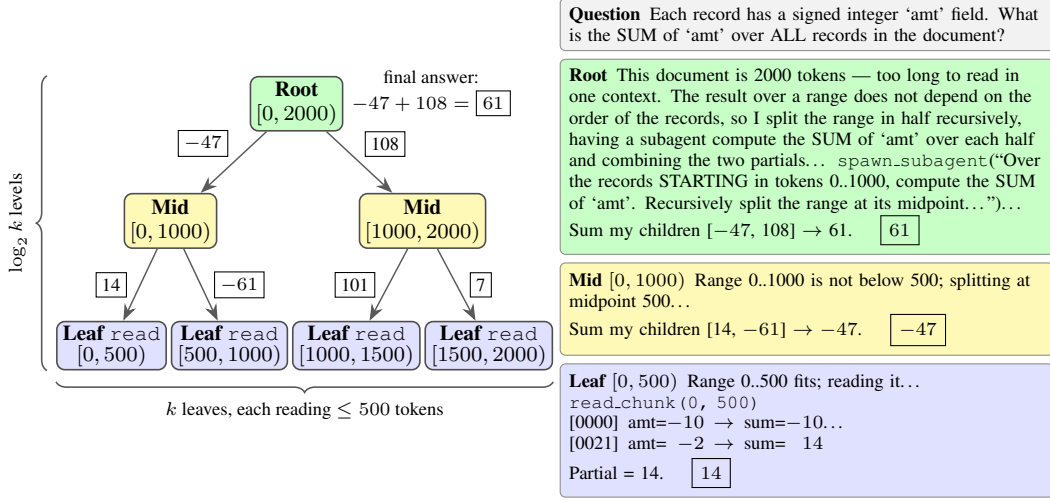


Figure 1: **Tree reduce**, for order-independent tasks (aggregation, retrieval). Non-leaf nodes delegate half their range to subagents. Leaves read one chunk and compute a leaf op which is aggregated up the tree.

<sup>4</sup>Opus 4.6 does appear to perform somewhat decently on our harness with unlimited context budget. Unfortunately, due to the massive subagent trees, running the full comparison without restriction is cost-prohibitive.

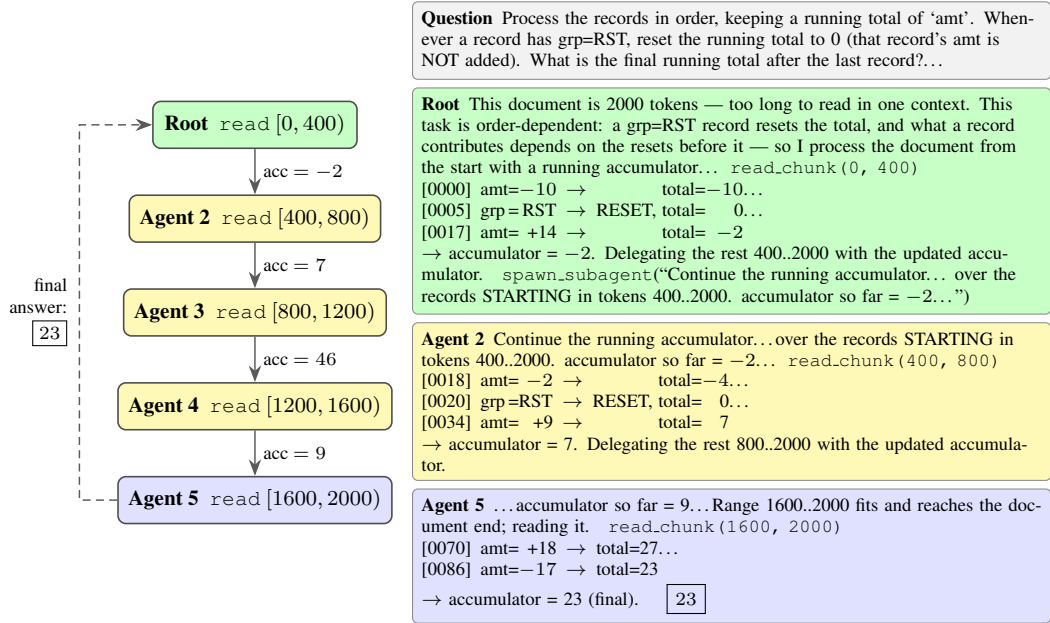


Figure 2: **Sequential fold**, for order-dependent tasks (variable tracking, running state). Each agent reads the next slice, updates the accumulator, and delegates the rest of the document to the next agent.



Figure 3: **Boundary handling**. A leaf/fold agent owns the data that *starts* in its range. It reads 200 additional tokens at a time until the last item in its range is not cut off (additional calls not pictured in Figures 1 and 2, for brevity).

We generate synthetic oracle demonstration data on a wide distribution of aggregation, retrieval, and sequential tasks. While we cover similar shapes to our evals, our training distribution uses independent questions and vastly more diverse examples. The technical challenge is to instill the necessary prior so the model knows how to take a question and delegate appropriate subtasks to agents from which the final answer can be determined.

For example, consider the training question “In how many sections are there STRICTLY more named than unnamed sentences?” (the prompt defines a *named* sentence as a sentence with a proper noun). To answer this question, each leaf must compute a tally of named and unnamed sentences per section in its assigned range. Thus, the root must recognize the necessary, often multi-dimensional, key space and dictate this and a recursive strategy to mid-level agents, who pass the instruction down

to other mid-level agents, until leaves compute their range and counts are aggregated back up the tree, culminating in a final judgement at the root.

Our training data is primarily record-based, but also includes real prose, dot-separated word lists, and haystacks with injected needles. We finetune Qwen3.6-35B-A3B, chosen as a capable small model with affordable compute, trained on the Tinker platform (Thinking Machines Lab, 2025a;b). For more details on our complete training data and training hyperparameters, see Appendices B and C, respectively.

We do consider an RL stage following SFT. Unfortunately, even with LLM-as-a-judge rewards on subagent trajectories, as described in Recursive Agent Optimization (Gandhi et al., 2026), the policy exhibits high-variance and latches onto simpler strategies that weaken overall performance. We suspect RL to be useful in learning recovery and strengthening generalization; however, to do this properly probably requires group-level baselines on subagent trajectories, to cut variance, and compute beyond the scope of this paper.

## 4 EXPERIMENTS

We test our finetune on RULER (Hsieh et al., 2024), the standard long context eval, and OOLONG-synth (Bertsch et al., 2025), a newer long context benchmark frontier models have not yet saturated, also used in Recursive Language Models (Zhang et al., 2025), Recursive Agent Optimization (Gandhi et al., 2026), and Recursive Agent Harnesses (Lumer et al., 2026). RULER is an extension of needle-in-a-haystack style tasks with variable tracking, word frequencies, and Q&A. OOLONG-synth composes classification tasks on ten datasets into long documents on which aggregation is queried.

We compare against base Qwen3.6-35B-A3B<sup>5</sup> and GPT-5.4. We chose GPT-5.4 as a selection from the best frontier models available upon Qwen3.6-35B-A3B release. While we do not eval against additional frontier models, due to compute requirements, we recommend the reader read the OOLONG paper for more comparisons (GPT-5 tops the chart) (Bertsch et al., 2025). While we do eval GPT-5.4 on RULER, this benchmark is largely saturated for frontier models. For more RULER reference points, consult model release cards.

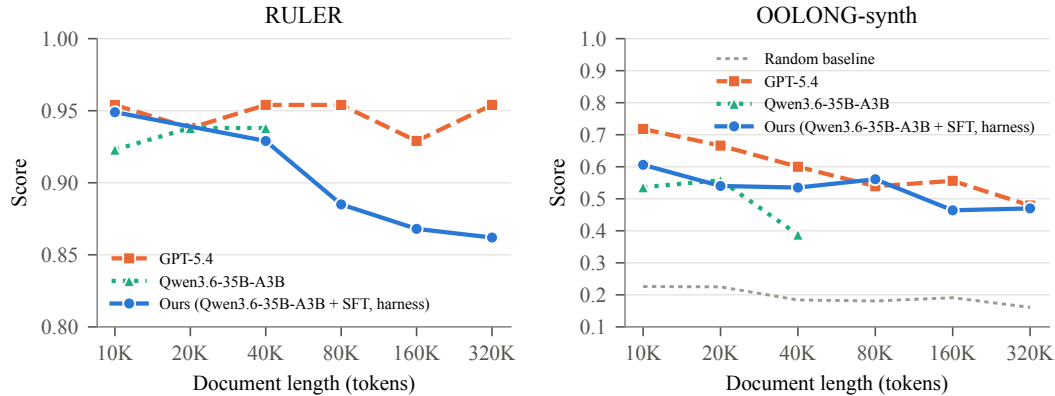


Figure 4: **Results on RULER and OOLONG-synth.** Our finetune runs in our harness with an 8K token per-agent context limit. GPT-5.4 (medium reasoning) and base Qwen3.6-35B-A3B are given the full document with the question (see Appendix A for details). Base Qwen and GPT-5.4 are served with 64K and 1M token context limits, respectively. The OOLONG random baseline follows Bertsch et al. (2025, §2.3), computed on the same problems.

For RULER, we fix a sample of 5 questions per problem type, for a total of 65 problems per document length. For OOLONG-synth, we fix a sample of 10 questions per family: counting, user, and

<sup>5</sup>While Qwen3.6-35B-A3B was trained with a 256K token context limit, we sample Qwen3.6-35B-A3B from Tinker, which caps the model at 64K token context.

temporal, each sampled from a different underlying classification dataset. We vendor generators for both benchmarks and generate our problems from scratch. We eval our finetune at temperature 0.2 (determined best empirically), GPT-5.4 at API default (required for reasoning mode), and base Qwen3.6-35B-A3B at temperature 0 (we found temperature 0 and 0.2 to perform similarly).

On RULER, our finetune performs worse than GPT-5.4, while remaining above 85%. On OOLONG-synth, our model starts worse than GPT-5.4, then matches its performance when document size exceeds 40K. At this length, native context performance begins to degrade, and a single model can be overwhelmed with the number of items requiring tracking within its context. On the other hand, agents in our harness operate in 8K token budgets, executing small pieces of an overall decomposition strategy, which scales gracefully to arbitrary lengths.

Per problem-family results are given in Table 1.

| RULER           | 10K   | 40K   | 80K   | 160K  | 320K               |          |       |       |       |       |       |       |
|-----------------|-------|-------|-------|-------|--------------------|----------|-------|-------|-------|-------|-------|-------|
| NIAH (8 tasks)  | 1.00  | 1.00  | 1.00  | 1.00  | 1.00               | OOLONG   | 10K   | 20K   | 40K   | 80K   | 160K  | 320K  |
| VT              | 1.00  | 1.00  | 0.92  | 0.84  | 0.70 <sup>†</sup>  | Counting | 0.45  | 0.42  | 0.42  | 0.41  | 0.41  | 0.40  |
| CWE             | 1.00  | 0.74  | 0.58  | 0.24  | 0.18               | User     | 0.86  | 0.72  | 0.80  | 0.73  | 0.61  | 0.60  |
| FWE             | 0.93  | 0.93  | 1.00  | 1.00  | 0.93               | Temporal | 0.51  | 0.48  | 0.39  | 0.54  | 0.37  | 0.41  |
| QA-1            | 0.60  | 0.80  | 0.60  | 0.60  | 0.60               | Mean     | 0.606 | 0.540 | 0.535 | 0.561 | 0.464 | 0.470 |
| QA-2            | 0.80  | 0.60  | 0.40  | 0.60  | 0.80               |          |       |       |       |       |       |       |
| Mean (13 tasks) | 0.949 | 0.929 | 0.885 | 0.868 | 0.862 <sup>†</sup> |          |       |       |       |       |       |       |

Table 1: **Results by task family.** **Left:** On RULER, our model chooses a lossy aggregation strategy for the common words “CWE” problem type, which degrades when common words are diluted by fillers as document length grows. GPT-5.4 scores 0.56 on CWE at doc size 320K. **Right:** On OOLONG-synth, leaf classification and merge errors compound at length. Distribution mismatch also arises at 160K+ document length (it sees  $\leq 14K$  doc size in training). <sup>†</sup>4 of 5 problems.

## 5 CONCLUSION

In this paper we show training on a bare bones agentic harness is sufficient for a small model with small context to perform as well as a frontier model with 1M token context on long-context evals. While we do not claim full generalization of our finetune, we also do not train on our evaluation distributions. The point of this paper is to show the power of agentic training in contrast to harness engineering, excessive native long context, and lossy compaction. As a side effect we also show the power of synthetic oracle SFT data generated by powerful frontier models (Opus 4.8, 5, 5.5, and Fable 5 and 5.1, in this case).

The thesis of the last decade of AI development is that specific model capability can be incorporated into fully-general models by training on a wider distribution. The methods of this paper can be incorporated into frontier model development, so such models exhibit improved agentic performance under context pressure, without wasting tokens. This is in contrast to an external harness which must be applied at runtime.

To develop our curriculum into a production recipe, the training distribution would need to be widened and a reinforcement learning stage would likely prove fruitful in supercharging agentic capabilities. We leave this as important future work, outside the scope of this paper or a small research project more generally.

## AI USE STATEMENT

In this work, we used generative AI tools to write all our code, administer experiments, create figures, search and analyze relevant literature, format references, generate synthetic data sets, propose or refine hypotheses/brainstorming, design or provide feedback on research methodology/experiments, clean and reformat data, support qualitative and thematic data analysis, and interpret results. We have not used generative AI tools to draft parts of this research paper, and the following topics are not applicable to this work: help develop theoretical models or conceptual frameworks, formulate mathematical claims, provide critical ingredients for proving mathematical

claims, assist in the writing of proofs, and assist with translation. We have reviewed our AI-assisted work, for example, by reviewing AI code and conducting manual audits of synthetic data. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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## A HARNESS AND PROMPTS

The full system prompt and tool descriptions are given in Figure 5. The full-document prompt, used for single shot baselines, is given in Figure 6. Benchmark text fields are supplied in Table 2. Harness responses are given in Table 3.

You are a long-document assistant. The document is  $\langle N \rangle$  tokens long; you cannot see it directly. Your own context window is  $\langle B \rangle$  tokens — the conversation (system prompt, user message, your responses, and all tool results) must fit in this budget or the episode ends.

*$\langle$ task context, if the benchmark provides one $\rangle$*

You have two tools:

- `'read.chunk(start, end)'`: read the document tokens in  $[start, end)$ .
- `'spawn.subagent(subtask)'`: delegate to a fresh-context copy of yourself (also  $\langle B \rangle$  tokens) with `'subtask'` as the user prompt. The same tools and system prompt as above are inherited by the subagent.

When you are confident in the final answer, emit it as `\boxed{value}` and stop.

### Tool descriptions (function-calling schema)

`read.chunk`: Read a slice of the document and return the decoded text of tokens  $[start, end)$ . The document has a fixed length (stated in the system prompt). Each call is capped at the chunk limit; for larger ranges, issue multiple reads or delegate to a subagent.

`start`: First token position to read (inclusive). `end`: Last token position to read (exclusive).

`spawn.subagent`: Spawn a fresh-context copy of yourself to solve `'subtask'`; returns the child's `\boxed{}` answer.

`subtask`: Sub-problem statement (free text) for the child to solve

Figure 5: **Agent system prompt**, shared by the root and every subagent, and the two tool descriptions.  $\langle N \rangle$  is the document length and  $\langle B \rangle$  the per-agent context budget. The root's user message is the benchmark question (Table 2); a subagent's user message is the subtask string its parent wrote.

### System

*$\langle$ task context, if the benchmark provides one $\rangle$*

You are given a complete document followed by a question about it. Read the document carefully and answer the question. Show your reasoning, then emit your final answer as `\boxed{value}` and stop.

### User

*$\langle$ full document text $\rangle$*

*$\langle$ question $\rangle$*

Figure 6: **Full-document prompt**, used for the single-shot baselines: the whole document is placed in one user message, with no tools and no budget prose. For RULER, whose template embeds the document mid-prompt, the document is instead placed back at the template's `{context}` position, reproducing RULER's own prompt.

| Benchmark    | System prompt task context                                                             | Root user message                                                                                                                       |
|--------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| OOLONG-synth | OOLONG’s dataset description, verbatim (its intro, the number of items, the label set) | OOLONG’s question, verbatim, + “Put your final answer inside \boxed{.”                                                                  |
| RULER        | None (empty)                                                                           | RULER’s task template, verbatim, with {context} replaced by a pointer to the document (below) + “Reply with the answer inside \boxed{.” |

Table 2: **Benchmark text fields.** The task context is placed in the system prompt, so every subagent inherits it without the parent restating it. For RULER, {context} in the template is replaced by: “[The relevant text is in a separate document accessible via the read\_chunk tool — see the system prompt for usage.] (Document length:  $\langle N \rangle$  tokens.)”

| Event                                 | Tool result returned to the calling agent (verbatim)             |
|---------------------------------------|------------------------------------------------------------------|
| <i>spawn_subagent</i>                 |                                                                  |
| Subagent ends with a boxed answer     | The content of its last closed \boxed{...}                       |
| Subagent exceeds its context budget   | agent overflowed context                                         |
| Subagent stops without a boxed answer | agent did not box an answer                                      |
| Tree node cap reached                 | Error: rollout node budget reached.<br>Solve directly.           |
| <i>read_chunk</i>                     |                                                                  |
| Valid range                           | The decoded text of tokens [start, end), clamped to the document |
| Empty range after clamping            | Empty range.                                                     |
| Non-integer arguments                 | Error: read_chunk requires integer 'start' and 'end'.            |

Table 3: **Harness responses.** An agent’s budget counts its whole rendered conversation (system prompt, tool schema, user message, its own turns and every tool result); each generation is capped at the remaining room. A subagent that answers returns only its boxed content, so its reads and reasoning never enter the parent’s context.

## B TRAINING DATA

A high-level description of our training tasks is given in Table 4.

| Task                                                                               | Strategy | $N$   | Example question / what the leaf does                                                         |
|------------------------------------------------------------------------------------|----------|-------|-----------------------------------------------------------------------------------------------|
| <i>Order-independent aggregation over structured records (one record per line)</i> |          |       |                                                                                               |
| synth_sum                                                                          | tree     | 10    | SUM of amt over all records                                                                   |
| synth_count                                                                        | tree     | 10    | how many records have flag=Y                                                                  |
| synth_max                                                                          | tree     | 10    | largest amt                                                                                   |
| synth_min                                                                          | tree     | 10    | smallest amt                                                                                  |
| synth_sumwhere                                                                     | tree     | 10    | SUM of amt over only the flag=Y records                                                       |
| synth_maxwhere                                                                     | tree     | 10    | largest amt among a flag-filtered subset                                                      |
| synth_mode                                                                         | tree     | 10    | most common grp (per-grp tally)                                                               |
| synth_distinct                                                                     | tree     | 10    | how many distinct grp values (set state)                                                      |
| synth_sumby                                                                        | tree     | 10    | which grp has the largest total amt                                                           |
| synth_count2                                                                       | tree     | 10    | count under a conjunction of two field conditions                                             |
| synth_diff                                                                         | tree     | 10    | SUM over flag=Y minus SUM over flag=N                                                         |
| synth_count_cmp                                                                    | tree     | 10    | how many records have amt below a threshold                                                   |
| synth_count_range                                                                  | tree     | 10    | how many records have amt in an interval                                                      |
| synth_filter_argmax                                                                | tree     | 15    | most / least common value among a filtered subset                                             |
| synth_2d                                                                           | tree     | 24    | two-level questions (e.g. for how many months is K3 the most common group)                    |
| <i>Sequential folds</i>                                                            |          |       |                                                                                               |
| synth_runreset                                                                     | fold     | 20    | running total that resets at grp=RST                                                          |
| synth_varchain                                                                     | fold     | 20    | final value of a variable under set B = A copies                                              |
| synth_peak                                                                         | fold     | 20    | highest value the running total reaches                                                       |
| synth_streak                                                                       | fold     | 20    | longest run of consecutive flag=Y records                                                     |
| synth_adjacent                                                                     | fold     | 20    | records whose amt exceeds the previous record's                                               |
| synth_first_exceed                                                                 | fold     | 20    | index where the running total first exceeds a threshold                                       |
| vt_novel                                                                           | fold     | 150   | variable-assignment chains hidden in prose                                                    |
| <i>Open-vocabulary frequency</i>                                                   |          |       |                                                                                               |
| synth_topk                                                                         | tree     | 120   | the $k$ most frequent words in a word list; partials are pruned word:count tallies            |
| <i>Multi-line records</i>                                                          |          |       |                                                                                               |
| long_records                                                                       | tree     | 120   | header + prose-body entries; counts, argmaxes, 2-D tallies, and word mentions in bodies       |
| <i>Labelling real text, then aggregating</i>                                       |          |       |                                                                                               |
| labeled_records                                                                    | tree     | 240   | items from 7 labelled datasets; the leaf judges each label, then counts / compares            |
| rule_label                                                                         | tree     | 40    | prose sentences labelled by a stated rule (e.g. ends with "?"), then aggregated               |
| realdoc_count                                                                      | tree     | 40    | occurrences of a word in a Gutenberg novel passage                                            |
| <i>Retrieval and QA in prose</i>                                                   |          |       |                                                                                               |
| niah_novel                                                                         | tree     | 60    | find the one sentence stating a key's magic value                                             |
| niah_multi                                                                         | tree     | 60    | collect several hidden facts, or a lookup instruction plus all facts, and resolve at the root |
| niah_bridge                                                                        | tree     | 80    | two-hop: fact 1 names who holds a role, fact 2 answers about that person                      |
| narrativeqa                                                                        | tree     | 30    | short-answer questions over book / script windows; model-executed leaf                        |
| Total                                                                              |          | 1,229 |                                                                                               |

Table 4: **Training tasks.**  $N$  is the number of problems per task. *Tree*: recursive halving, with each parent merging its two children's partials (for retrieval, the half that finds the answer wins). *Fold*: one agent reads a slice and hands the rest of the document and its accumulator to the next. Every agent in each tree is a training example.

We upweight root gradient mass 4x and positive narrativeqa leaves 2x, downsampling negative leaves by half. Each task has prompt jitter to train a wider distribution. We also jitter the system prompt. We train with context budgets 3K (50%), 5K (15%), 8K (13%), 10K (11%), and 12K (11%), and document length between 4K and 14K tokens, with a distribution so document length is typically larger than context budget.

## C TRAINING DETAILS

Our training runs used rank 32 LoRA on Qwen3.6-35B-A3B, chat renderer qwen3\_5\_disable\_thinking, cross-entropy loss on assistant tokens only. Small batch training of size 8-64 (you pay per token on Tinker, and small batch training is generally better-behaved per token, particularly on LoRAs (Marek et al., 2025; Schulman & Thinking Machines Lab, 2025; Masters & Luschei, 2018; Keskar et al., 2017)). We used pure Adam optimizer, no weight decay, no gradient clipping,  $\epsilon = 1\text{e-}12$ ,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.95$ , with constant learning rate between  $5\text{e-}6$  and  $1\text{e-}5$ , single epoch training.